ATIVIDADE DA AULA 5 - Banco de Dados - Relacional (2º Semestre)

ALUNA: Nadla Fernandes Ferreira

RA: 2581392613046

CÓDIGO SQL:

```
CREATE TABLE TipoEvento(
idTipoEvento INT PRIMARY KEY,
nome VARCHAR(100),
descricao TEXT
);
CREATE TABLE Localizacao(
idLocalizacao INT PRIMARY KEY,
latitude DECIMAL(9,6),
longitude DECIMAL(9,6),
cidade VARCHAR(100),
estado VARCHAR(2)
);
CREATE TABLE Usuario(
idUsuario INT PRIMARY KEY,
nome VARCHAR(100),
email VARCHAR(150),
senhaHash VARCHAR(255)
);
CREATE TABLE Evento(
idEvento INT PRIMARY KEY,
titulo VARCHAR(150),
descricao TEXT,
dataHora TIMESTAMP,
status VARCHAR(50),
idTipoEvento INT,
```

```
idLocalizacao INT,
FOREIGN KEY (idTipoEvento) REFERENCES TipoEvento(idTipoEvento),
FOREIGN KEY (idLocalizacao) REFERENCES Localizacao(idLocalizacao)
);
CREATE TABLE Relato(
idRelato INT PRIMARY KEY,
texto TEXT,
dataHora TIMESTAMP,
idEvento INT,
idUsuario INT,
FOREIGN KEY (idEvento) REFERENCES Evento(idEvento),
FOREIGN KEY (idUsuario) REFERENCES Usuario(idUsuario)
);
CREATE TABLE Alerta(
idAlerta INT PRIMARY KEY,
mensagem TEXT,
dataHora TIMESTAMP,
nivel VARCHAR(20),
idEvento INT,
FOREIGN KEY (idEvento) REFERENCES Evento(idEvento)
);
CREATE TABLE HistoricoEvento(
idHistorico INT PRIMARY KEY,
idEvento INT,
statusAnterior VARCHAR(50),
statusNovo VARCHAR(50),
dataAlteracao TIMESTAMP,
FOREIGN KEY (idEvento) REFERENCES Evento(idEvento)
);
INSERT INTO TipoEvento VALUES
(1,'Queimada','Incêndio de grandes proporções em áreas urbanas ou rurais.'),
```

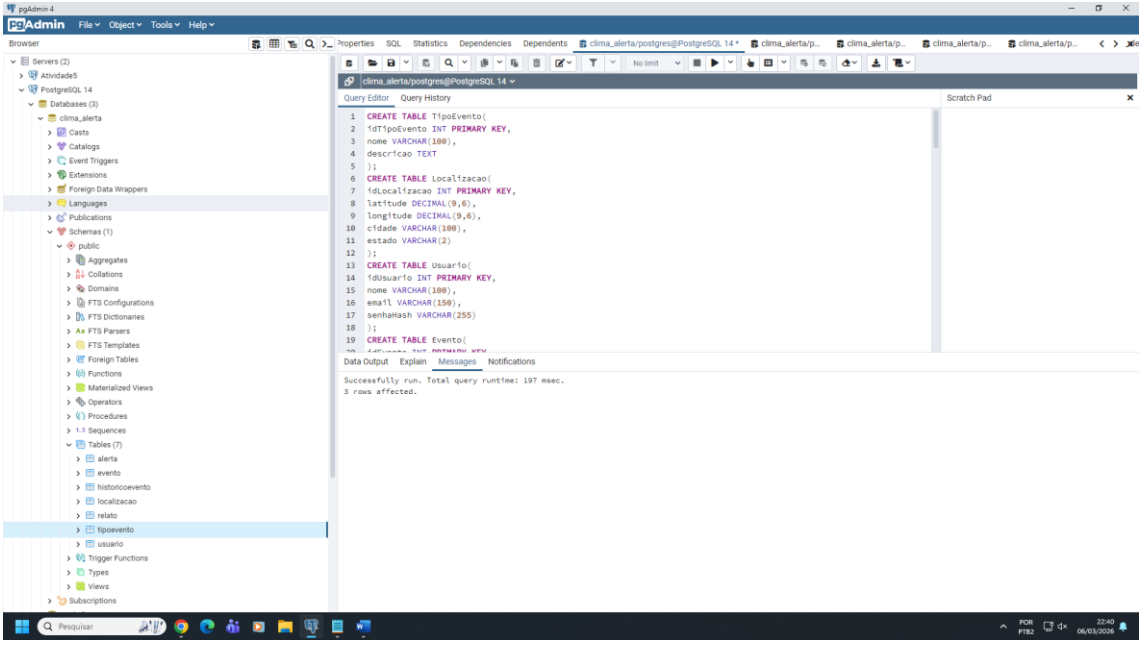
```
(4,'Chuva intensa', 'Chuva forte em área urbana ou rural'),
(3, 'Deslizamento de terra', 'Movimento de terra ou rochas');
INSERT INTO Localizacao VALUES
(5,-23.305,-45.965,'Jacareí','SP'),
(4,-45.904,-32.245,'Jacareí','SP'),
(6,-67.345,-54.342,'Jacareí','SP');
INSERT INTO Usuario VALUES
(2,'Maria Oliveira','maria.oliveira@email.com','hash123'),
(3,'João Augusto', 'jaugusto@email.com', 'hash456'),
(1, 'Maria Rita', 'marita@email.com','hash789');
INSERT INTO Evento VALUES
(1,
'Queimada em área de preservação',
'Fogo se alastrando na mata próxima à represa.',
'2025-08-15 14:35:00',
'Ativo',
1,
5
),
(2,
'Chuva intensa',
'Chuva intensa em áreas urbanas ou rurais pode haver risco de alagamento',
'2025-07-12 12:23:00',
'ativo',
4,
4
),
(3,
'Deslizamento de terra',
'Deslizamento de terra em encostas, pode apresentar riscos estruturais e a vida',
'2025-09-10 10:20:00',
```

```
'ativo',
4,
5
);
INSERT INTO Relato VALUES
(1,
'Fumaça intensa e chamas visíveis a partir da rodovia.',
'2025-08-15 15:10:00',
1,
2
);
INSERT INTO Alerta VALUES
(1,
'Evacuação imediata da área próxima à represa.',
'2025-08-15 15:20:00',
'Crítico',
1
);
SELECT nome, email
FROM Usuario;

SELECT titulo,descricao
FROM Evento;

SELECT titulo, dataHora
FROM Evento
WHERE DATE_PART('year', dataHora)= 2025;

SELECT cidade, estado
FROM Localizacao
WHERE estado = 'SP'
```


pgAdmin 4
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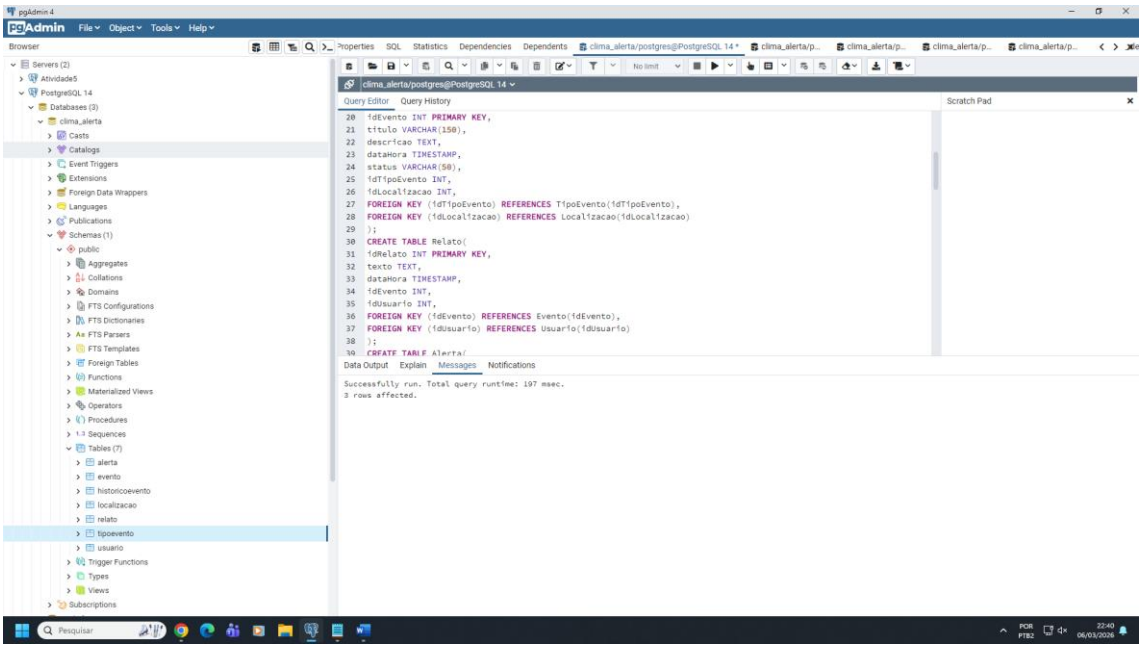
Query Editor Query History
Scratch Pad
CREATE TABLE TipoEvento(
idTipoEvento INT PRIMARY KEY,
nome VARCHAR(100),
descricao TEXT
);
CREATE TABLE Localizacao(
idLocalizacao INT PRIMARY KEY,
latitude DECIMAL(9,6),
longitude DECIMAL(9,6),
cidade VARCHAR(100),
estado VARCHAR(2)
);
CREATE TABLE Usuario(
idUsuario INT PRIMARY KEY,
nome VARCHAR(100),
email VARCHAR(150),
senhaHash VARCHAR(255)
);
CREATE TABLE Evento(
Data Output Explain Messages Notifications
Successfully run. Total query runtime: 197 msec.
3 rows affected.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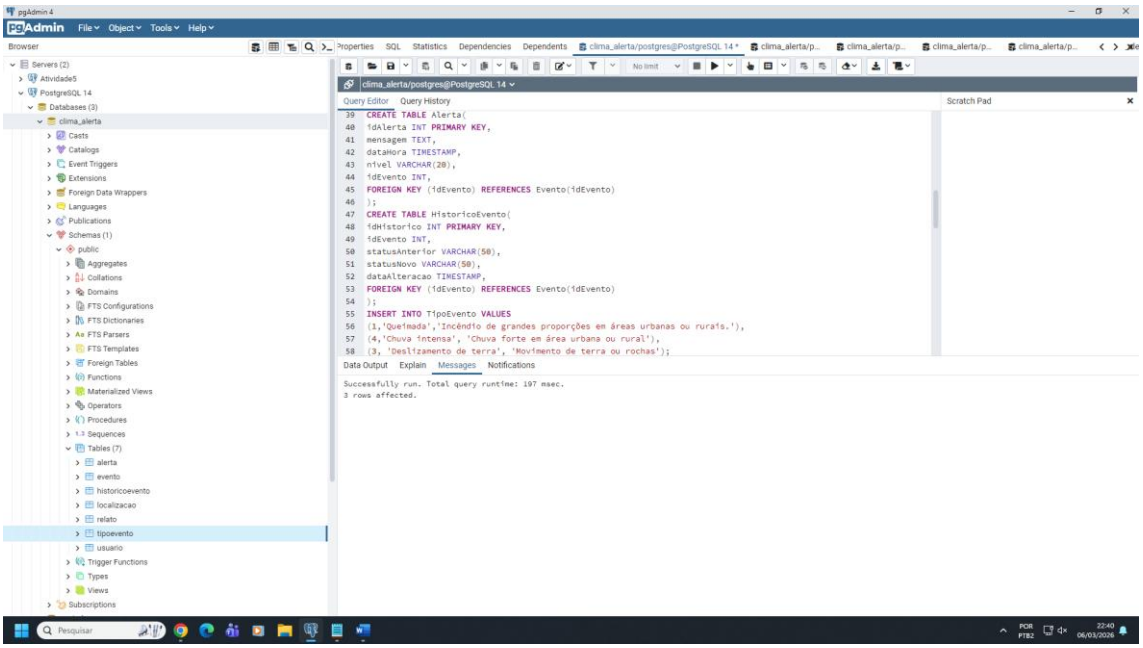
Query Editor Query History
Scratch Pad
idEvento INT PRIMARY KEY,
titulo VARCHAR(150),
descricao TEXT,
dataHora TIMESTAMP,
status VARCHAR(50),
idTipoEvento INT,
idLocalizacao INT,
FOREIGN KEY (idTipoEvento) REFERENCES TipoEvento(idTipoEvento),
FOREIGN KEY (idLocalizacao) REFERENCES Localizacao(idLocalizacao)
);
CREATE TABLE Relato(
idRelato INT PRIMARY KEY,
texto TEXT,
dataHora TIMESTAMP,
idEvento INT,
idUsuario INT,
FOREIGN KEY (idEvento) REFERENCES Evento(idEvento),
FOREIGN KEY (idUsuario) REFERENCES Usuario(idUsuario)
);
Data Output Explain Messages Notifications
Successfully run. Total query runtime: 197 msec.
3 rows affected.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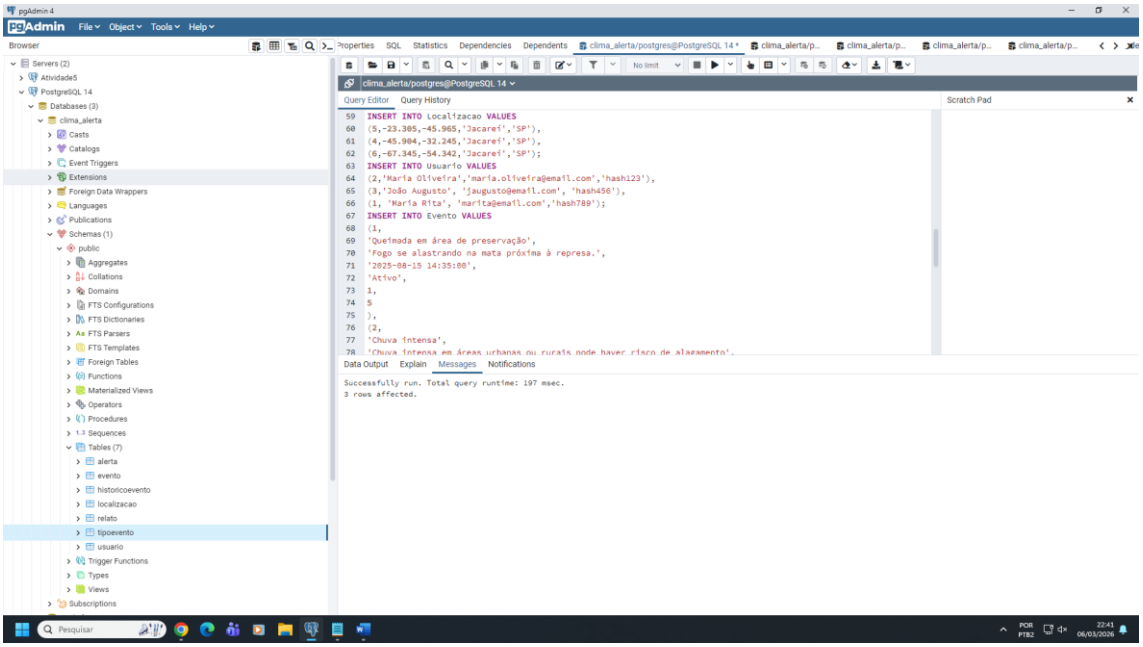
Properties SQL Statistics Dependencies Dependents
clima_alerta/postgres@PostgreSQL 14 *
clima_alerta/postgres@PostgreSQL 14
Query Editor Query History
Scratch Pad
39 CREATE TABLE Alerta(
40 idAlerta INT PRIMARY KEY,
41 mensagem TEXT,
42 dataHora TIMESTAMP,
43 nivel VARCHAR(20),
44 idEvento INT,
45 FOREIGN KEY (idEvento) REFERENCES Evento(idEvento)
46);
47 CREATE TABLE HistoricoEvento(
48 idHistorico INT PRIMARY KEY,
49 idEvento INT,
50 statusAnterior VARCHAR(50),
51 statusNovo VARCHAR(50),
52 dataAlteracao TIMESTAMP,
53 FOREIGN KEY (idEvento) REFERENCES Evento(idEvento)
54);
55 INSERT INTO TipoEvento VALUES
56 (1,'Queimada','Incêndio de grandes proporções em áreas urbanas ou rurais.'),
57 (4,'Chuva intensa', 'Chuva forte em área urbana ou rural'),
58 (3, 'Deslizamento de terra', 'Movimento de terra ou rochas');
Data Output Explain Messages Notifications
Successfully run. Total query runtime: 197 msec.
3 rows affected.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Properties SQL Statistics Dependencies Dependents
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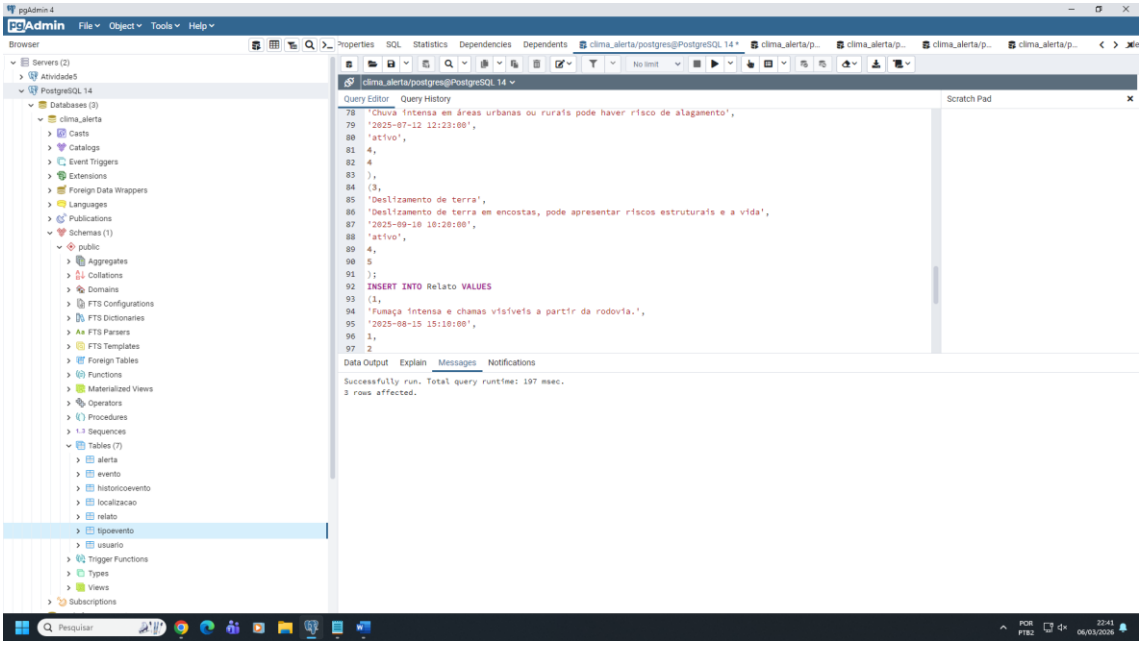
clima_alerta/postgres@PostgreSQL 14
Query Editor Query History
Scratch Pad
59 INSERT INTO Localizacao VALUES
60 (5,-23.305,-45.965,'Jacareí','SP'),
61 (4,-45.904,-32.245,'Jacareí','SP'),
62 (6,-67.345,-54.342,'Jacareí','SP');
63 INSERT INTO Usuario VALUES
64 (2,'Maria Oliveira','maria.oliveira@email.com','hash123'),
65 (3,'João Augusto', 'jaugusto@email.com', 'hash456'),
66 (1, 'Maria Rita', 'marita@email.com','hash789');
67 INSERT INTO Evento VALUES
68 (1,
69 'Queimada em área de preservação',
70 'Fogo se alastrando na mata próxima à represa.',
71 '2025-08-15 14:35:00',
72 'Ativo',
73 1,
74 5
75),
76 (2,
77 'Chuva intensa',
78 'Chuva intensa em áreas urbanas ou rurais pode haver risco de alagamento',
Data Output Explain Messages Notifications
Successfully run. Total query runtime: 197 msec.
3 rows affected.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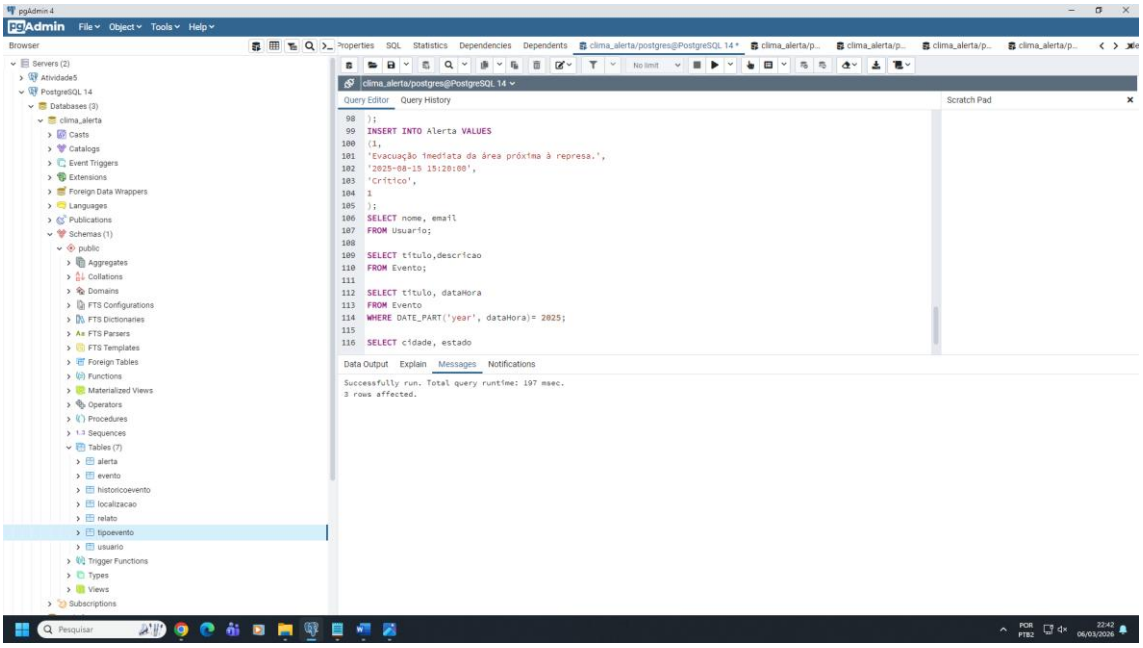
Query Editor Query History
Scratch Pad
78 'Chuva intensa em áreas urbanas ou rurais pode haver risco de alagamento',
79 '2025-07-12 12:23:00',
80 'ativo',
81 4,
82 4
83),
84 (3,
85 'Deslizamento de terra',
86 'Deslizamento de terra em encostas, pode apresentar riscos estruturais e a vida',
87 '2025-09-10 10:20:00',
88 'ativo',
89 4,
90 5
91);
92 INSERT INTO Relato VALUES
93 (1,
94 'Fumaça intensa e chamas visíveis a partir da rodovia.',
95 '2025-08-15 15:10:00',
96 1,
97 2
Data Output Explain Messages Notifications
Successfully run. Total query runtime: 197 msec.
3 rows affected.
Pesquisar
POR PTB2
22:41 06/03/2026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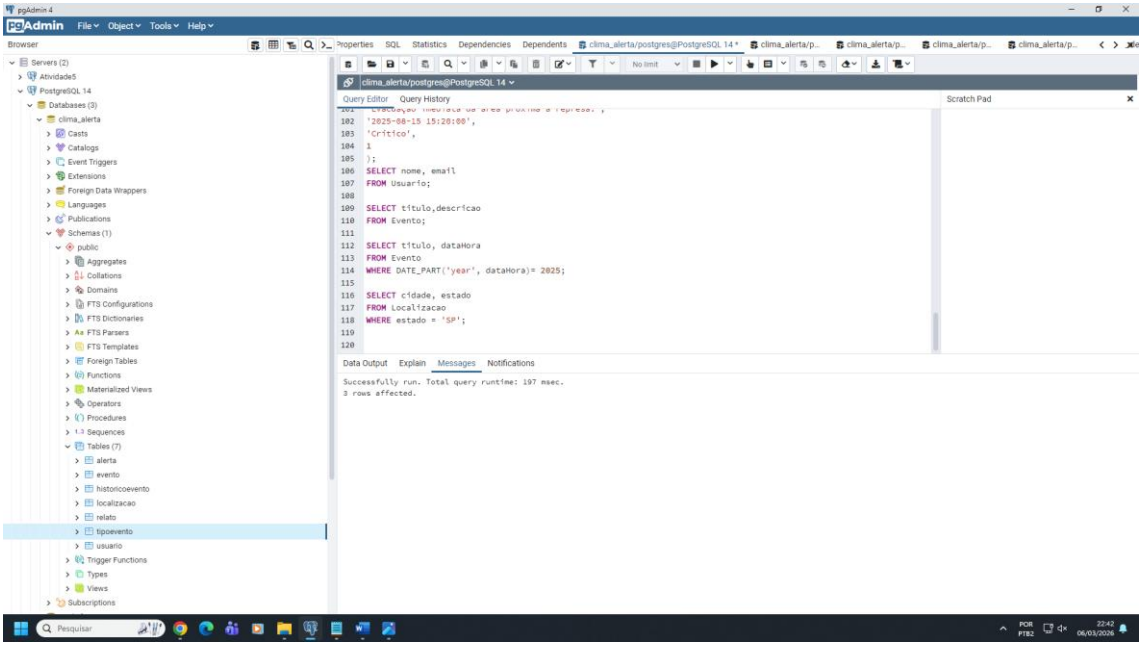
98);
99 INSERT INTO Alerta VALUES
100 (1,
101 'Evacuação imediata da área próxima à represa.',
102 '2025-08-15 15:20:00',
103 'Crítico',
104 1
105);
106 SELECT nome, email
107 FROM Usuario;
108
109 SELECT título,descricao
110 FROM Evento;
111
112 SELECT título, dataHora
113 FROM Evento
114 WHERE DATE_PART('year', dataHora)= 2025;
115
116 SELECT cidade, estado
Data Output Explain Messages Notifications
Successfully run. Total query runtime: 197 msec.
3 rows affected.
Pesquisar
POR PTB2
22:42 06/03/2026


pgAdmin 4
pgAdmin File Object Tools Help
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Properties SQL Statistics Dependencies Dependents clima_alerta/postgres@PostgreSQL 14 * clima_alerta/p... clima_alerta/p... clima_alerta/p... clima_alerta/p...
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clima_alerta/postgres@PostgreSQL 14
Query Editor Query History
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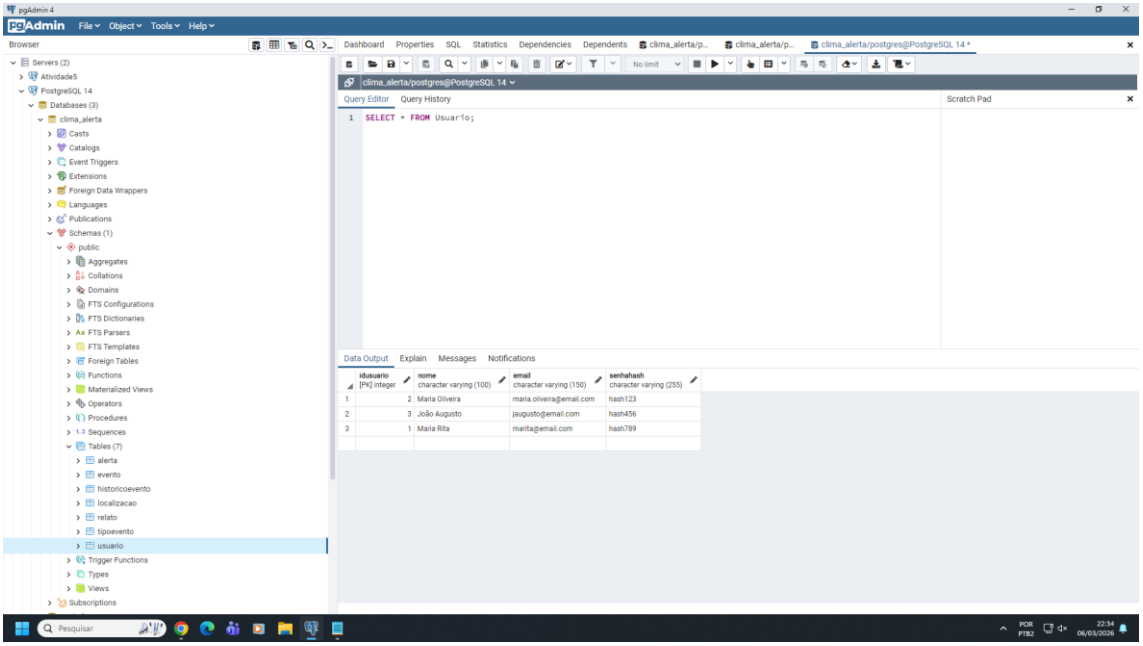
'2025-08-15 15:20:00',
'Crítico',
1
);
SELECT nome, email
FROM Usuario;
SELECT titulo,descricao
FROM Evento;
SELECT titulo, dataHora
FROM Evento
WHERE DATE_PART('year', dataHora)= 2025;
SELECT cidade, estado
FROM Localizacao
WHERE estado = 'SP';
Data Output Explain Messages Notifications
Successfully run. Total query runtime: 197 msec.
3 rows affected.
Pesquisar
POR PTB2
22:42 06/03/2026


pgAdmin 4
pgAdmin File Object Tools Help
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Dashboard Properties SQL Statistics Dependencies Dependents clima_alerta/p... clima_alerta/p... clima_alerta/postgres@PostgreSQL 14 *
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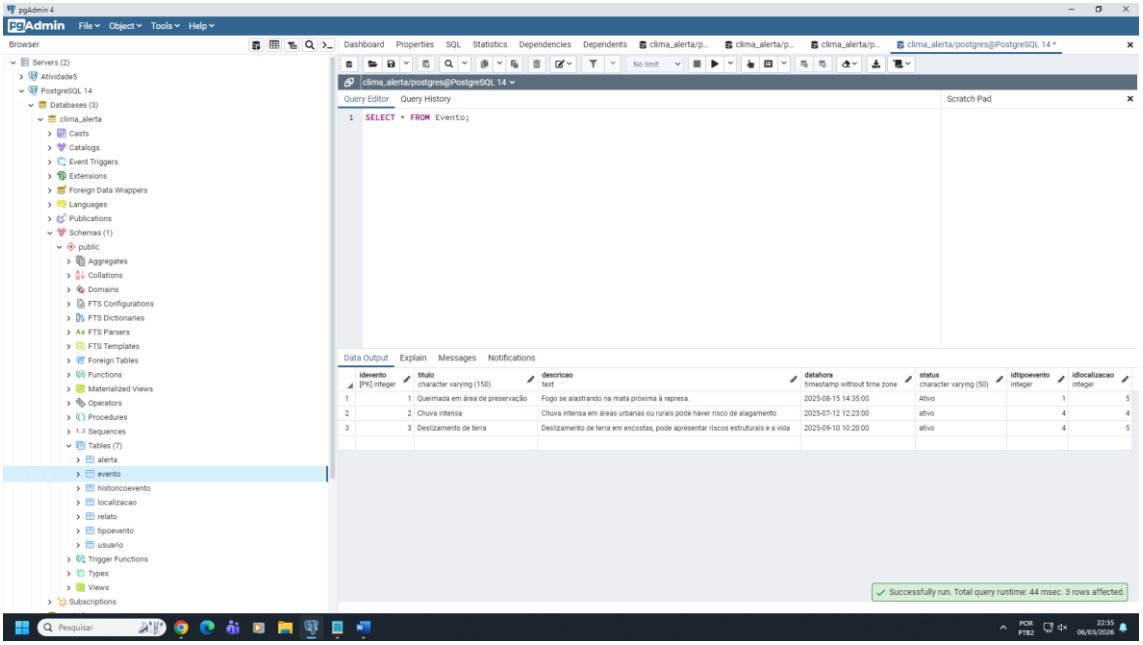
No limit
clima_alerta/postgres@PostgreSQL 14
Query Editor Query History
Scratch Pad
SELECT * FROM Usuario;
Data Output Explain Messages Notifications
idusuario [PK] integer
nome character varying (100)
email character varying (150)
senhahash character varying (255)
1 2 Maria Oliveira maria.oliveira@email.com hash123
2 3 João Augusto jaugusto@email.com hash456
3 1 Maria Rita marita@email.com hash789
Pesquisar
POR PTB2
22:34 06/03/2026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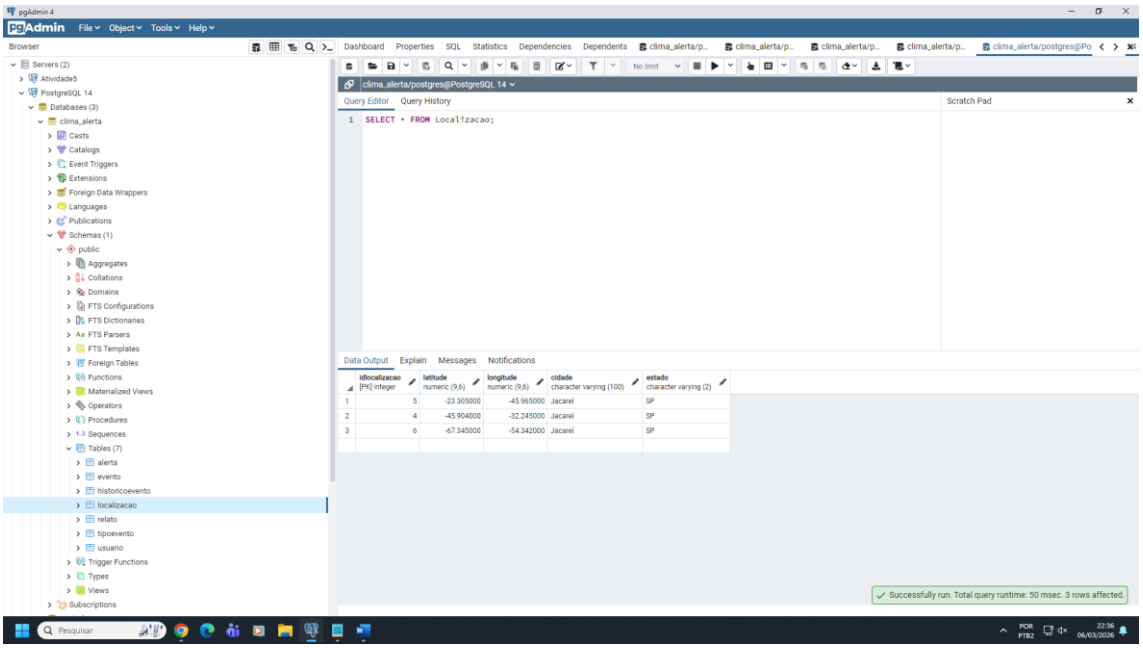
Query Editor Query History
Scratch Pad
SELECT * FROM Evento;
Data Output Explain Messages Notifications
idevento [PK] integer
titulo character varying (150)
descricao text
datahora timestamp without time zone
status character varying (50)
idtipoevento integer
idlocalizacao integer
1 1 Queimada em área de preservação Fogo se alastrando na mata próxima à represa. 2025-08-15 14:35:00 Ativo 1 5
2 2 Chuva intensa Chuva intensa em áreas urbanas ou rurais pode haver risco de alagamento 2025-07-12 12:23:00 ativo 4 4
3 3 Deslizamento de terra Deslizamento de terra em encostas, pode apresentar riscos estruturais e a vida 2025-09-10 10:20:00 ativo 4 5
Successfully run. Total query runtime: 44 msec. 3 rows affected.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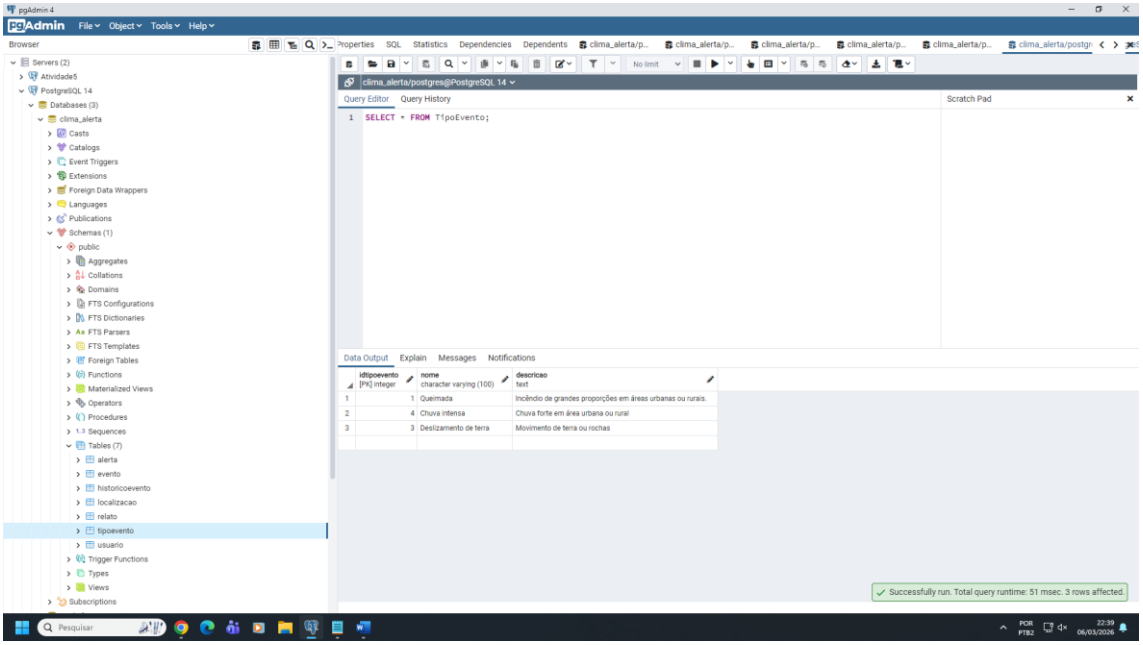
clima_alerta/postgres@PostgreSQL 14
Query Editor Query History
Scratch Pad
SELECT * FROM Localizacao;
Data Output Explain Messages Notifications
idlocalizacao [PK] integer
latitude numeric (9,6)
longitude numeric (9,6)
cidade character varying (100)
estado character varying (2)
1 5 -23.305000 -45.965000 Jacareí SP
2 4 -45.904000 -32.245000 Jacareí SP
3 6 -67.345000 -54.342000 Jacareí SP
Successfully run. Total query runtime: 50 msec. 3 rows affected.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Query Editor Query History
Scratch Pad
SELECT * FROM TipoEvento;
Data Output Explain Messages Notifications
idtipoevento [PK] integer
nome character varying (100)
descricao text
1 1 Queimada Incêndio de grandes proporções em áreas urbanas ou rurais.
2 4 Chuva intensa Chuva forte em área urbana ou rural
3 3 Deslizamento de terra Movimento de terra ou rochas
Successfully run. Total query runtime: 51 msec. 3 rows affected.